La Scuola d'Italia / Dr. Huson / IB Math 12
27 October 2026

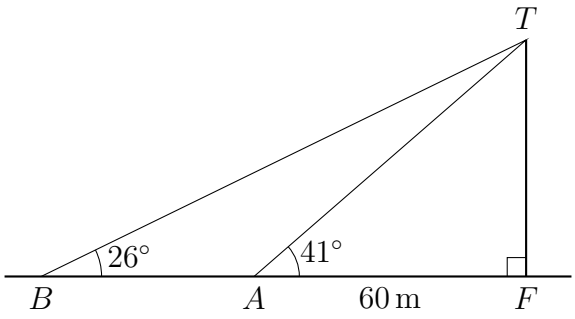
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2.5 Pre-Quiz: Trigonometry and Statistics

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \qquad c^2 = a^2 + b^2 - 2ab\cos C \qquad \text{Area} = \frac{1}{2}ab\sin C$$

A calculator is allowed. Give answers correct to three significant figures, unless they are exact. When a question gives an angle in radians, give angles in radians. Diagrams are not to scale.

1. A vertical cliff FT has its foot F on horizontal ground. Point A is on the ground, 60 m from F. The angle of elevation of the top T from A is 41°. Point B is on the ground, further from the cliff on the line FA, and the angle of elevation of T from B is 26°.

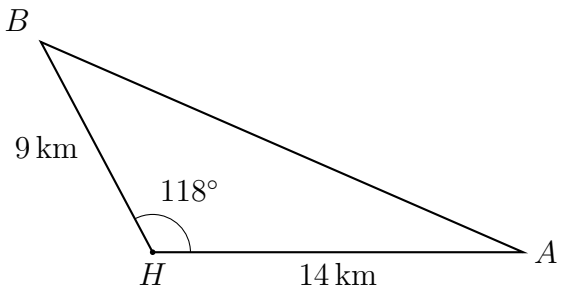


   (a) Find the height of the cliff.

   (b) Find the distance AB.

   (c) Find the distance BT.

2. Two boats leave a harbour H. Boat A sails 14 km in a straight line and boat B sails 9 km in a straight line. The angle between their paths is $A\hat{H}B = 118°$.



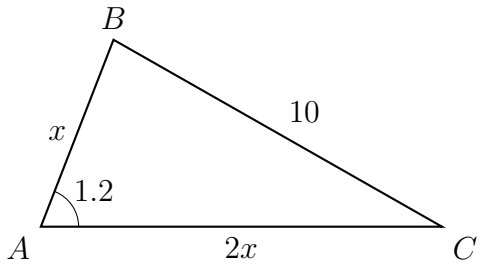
(a) Find the distance between the two boats.

(b) Find $H\hat{A}B$.

(c) Find the area of triangle HAB.

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3. The following diagram shows triangle ABC, with $AB = x$ cm, $AC = 2x$ cm, $BC = 10$ cm and $B\hat{A}C = 1.2$ radians.

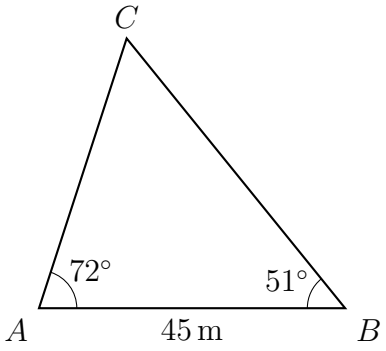


(a) Find the value of x.

(b) Find the area of triangle ABC.

(c) Find $A\hat{C}B$.

4. A triangular plot of land ABC has $AB = 45\,\text{m}$, $B\hat{A}C = 72°$ and $A\hat{B}C = 51°$.



(a) Write down $A\hat{C}B$.

(b) Find AC.

(c) Find the area of the plot.

A second triangular plot has two sides of length 30 m and 40 m. Its area is $500\,\text{m}^2$, and the angle between these two sides is obtuse.

(d) Find the angle between these two sides.

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5. Fourteen students recorded the number of minutes they spent on homework one evening.

34, 41, 28, 45, 39, 54, 31, 37, 43, 36, 40, 33, 85, 38

(a) Write down the median, the lower quartile Q_1 and the upper quartile Q_3.

(b) Determine whether 54 and 85 are outliers. Show your working.

(c) Find the mean and the standard deviation of the data.

(d) The value 85 was a recording error and is removed. Find the mean of the remaining 13 values.

6. The table shows the number of hours, x, that eight students studied for a test, and their scores, y.

Hours, x	2	3	5	6	8	9	11	12
Score, y	52	55	61	60	70	73	79	84

(a) Find Pearson's product-moment correlation coefficient, r, and describe the correlation.

(b) Write down the regression line of y on x, in the form $y = ax + b$.

(c) Use your regression line to estimate the score of a student who studied for 10 hours.

(d) Explain why the regression line should not be used to estimate the score of a student who studied for 25 hours.